Lei Shen

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EDUCATION

- **Wuhan University**, Wuhan, Hubei, China — Sep 2023 – Present
  M.S. in Medicine in Surgery, GPA: 3.71/4.0
  Honors: National Scholarship for Graduate Students for two years (2023 & 2024)
- **Wuhan University**, Wuhan, Hubei, China — Sep 2018 – May 2023
  B.S. in Clinical Medicine, GPA: 3.58/4.0
  Honors: Outstanding Graduate of Wuhan University (2023), Lei Jun Scholarship of Wuhan University (2022)

RELEVANT COURSEWORKS

- **Graduate-level:** Evidence-Based Clinical Medicine (A), Introduction to Clinical Research (A), Guided Reading of Medical Literature (A), Academic Ethics and Regulations (A), Surgery (B), Medical Professionalism (A), etc.
- **Undergraduate-level:** Basic Medicine Comprehensive Experiment (A), Clinical Pathophysiology and Therapeutics (A), Neuroscience (B), Surgical Internship (A), Internal Medicine Internship (A), etc.

PUBLICATIONS

[1] **Lei Shen*, Zhourui Zhang, Pengfei Wu, et al. "Mechanistic insight into glioma through spatially multidimensional proteomics". In: *Science Advances* 10.7 (2024), eadk1721. doi: 10.1126/sciadv.adk1721.**

[2] Jingyi Yang, **Lei Shen***, Yuankun Cai, et al. "The Role of Coagulation-Related Genes in Glioblastoma: A Comprehensive Analysis of the Tumor Microenvironment, Prognosis, and Treatment". In: *Biochemical Genetics* (Mar. 2025). doi: 10.1007/s10528-025-11086-3.

[3] Jingyi Yang, **Lei Shen***, Jiabin Zhou, et al. "A Novel Mitochondrial-Related Gene Signature for the Prediction of Prognosis and Therapeutic Efficacy in Lower-Grade Glioma". In: *Biochemical Genetics* (Oct. 2024). doi: 10.1007/s10528-024-10928-w.

[4] **Lei Shen*** and Nan Xiong. "Letter: Evaluation of 2 Surgical Techniques—Transposition Versus Interposition Microvascular Decompression for Hemifacial Spasm: A Systematic Review of 19 437 Patients". In: *Neurosurgery* (May 2025). Epub ahead of print. doi: 10.1227/neu.0000000000003496.

[5] **Lei Shen***, Mengyang Wang, Jingwei Zhao, et al. "Study on the relationship between obesity and complications of Pediatric Epilepsy surgery". In: *BMC Pediatrics* 23.1 (Mar. 2023), p. 142. doi: 10.1186/s12887-023-03948-9.

[6] **Lei Shen***, Jingyi Yang, Runqi Cheng, et al. "Bridge-layered decompression technique for vertebral artery-involved hemifacial spasm: technical note". In: *BMC Surgery* 24.1 (May 2024), p. 154. doi: 10.1186/s12893-024-02415-1.

[7] Keyu Chen, **Lei Shen***, Jingyi Yang, et al. "A nomogram based on clinical multivariate factors predicts delayed cure after microvascular decompression for hemifacial spasm". In: *Neurosurgical Review* 47.1 (Jan. 2024), p. 44. doi: 10.1007/s10143-024-02284-5.

[8] **Lei Shen***, Huan Liu, Xue Liu, et al. "An adolescent patient with anti-N-methyl-D-aspartate receptor encephalitis with motor aphasia as the first symptom and complicated by peripheral nerve damage: A case report and literature review". In: *Medicine (Baltimore)* 104.19 (May 2025), e42436. doi: 10.1097/MD.0000000000042436.

[9] Jianmei Yang, **Lei Shen***, Jingyi Yang, et al. "Complement and coagulation cascades are associated with prognosis and the immune microenvironment of lower-grade glioma". In: *Translational Cancer Research* 13.1 (Jan. 2024), pp. 112–136. doi: 10.21037/tcr-23-906.

SELECTED PROJECTS

Project 1: Functional mechanism of PEPD in glioblastoma (Master thesis) Mar 2024 - Present
Location: Wuhan University, Xiong's Lab; Advisor: Prof. Nanxiang Xiong

Building on previous proteomic discoveries, I focused my Master's thesis on the role of PEPD in glioblastoma (GBM). I demonstrated that PEPD promotes GBM progression by regulating alternative splicing through RBM7-mediated Insig1 signaling, thereby proposing a novel mechanism for GBM malignancy.

- Independently reviewed literature, formulated hypotheses, designed and carried out research projects.
- Validated the PEPD–ZFP36–Insig1 axis using RNA-seq, qPCR, Western blot, Co-IP, RIP.

Project 2: Spatially multidimensional proteomics of glioma Sep 2021 - Mar 2024
Location: Wuhan University, Xiong's Lab; Advisor: Prof. Nanxiang Xiong

As a core member, I contributed to a multi-cohort proteomic study that uncovered novel molecular mechanisms in diffuse glioma. This collaborative work was published in *Science Advances* as the first-author paper titled "Mechanistic insight into glioma through spatially multidimensional proteomics".

- Performed proteomic sample preparation and downstream data analysis.
- Contributed substantially to data visualization and manuscript writing.

Project 3: The role of Cuproptosis in glioma progression Mar 2022 - Jun 2023
Location: Wuhan University, Xiong's Lab; Advisor: Prof. Nanxiang Xiong

Cuproptosis is a novel type of programmed cell death, discovered in 2022. I led a *National Training Program of Innovation and Entrepreneurship* to investigate whether cuproptosis regulates glioma progression. Our findings revealed that activation of cuproptosis suppressed glioma development, suggesting a potential therapeutic value.

- Applied bioinformatics pipelines to analyze glioma-related datasets for cuproptosis signatures.
- Conducted *in vitro* cellular and molecular experiments to validate functional roles of cuproptosis.

RELEVANT TECHNICAL SKILLS

Wet-lab skills:

- Cell culture (staining, scratch assay, colony formation, CCK8, Transwell).
- Molecular techniques (qPCR, Western blot, Immunohistochemistry, ELISA, Co-IP, ChIP,RIP).
- Basic animal handling, sampling, and behavioral testing.

Dry-lab skills: R, Python, bioinformatic analysis for bulk and single-cell RNA-seq, proteomics, metabolomics.

Clinical skills: Lumbar puncture, continuous lumbar subarachnoid drainage, and basic surgical skills.

Languages: Chinese (native), English (proficient)

LEADERSHIP & COMMUNITY SERVICE

Academic Peer Review Service Apr 2024 - Present
Serve as a reviewer for the Heliyon journal. Reviewed 2 papers.

Team Leader, Undergraduate Innovation Projects Sep 2022 - Sep 2024
Led 6 undergraduate students to complete two *National Training Program of Innovation and Entrepreneurship* for Undergraduates, providing mentorship on research design, data analysis, and project execution.

Student Committee Member, School Student Council Sep 2022 - June 2023
Coordinated with faculty to organize summer social practices, school sports events, and other student activities, ensuring smooth execution and student engagement.

Volunteer Coordinator, University Student Volunteer Association Sep 2019 - Sep 2021
Organized and led 4 volunteer initiatives, including HIV/AIDS prevention campaigns and first-aid education programs, engaging 400+ students and 2 local communities.

Class President, Clinical Medicine Undergraduate Program Sep 2018 - Jun 2020
Led a class of 23 students, organized academic and social activities to promote collaboration and cohesion among classmates, and coordinated with faculty on class matters.